UNIVERSITY OF ILLINOIS
AT URBANA-CHAMPAIGN

Cutting and tearing in highly deformable and tough hydrogels

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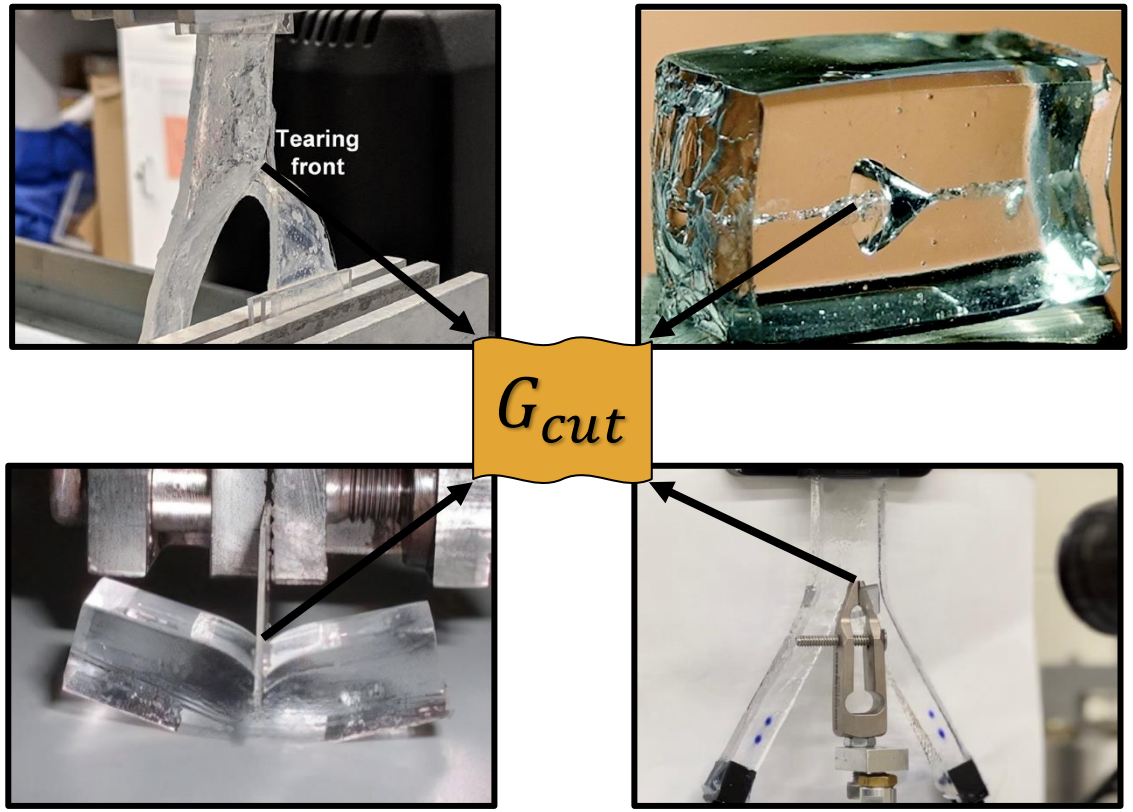
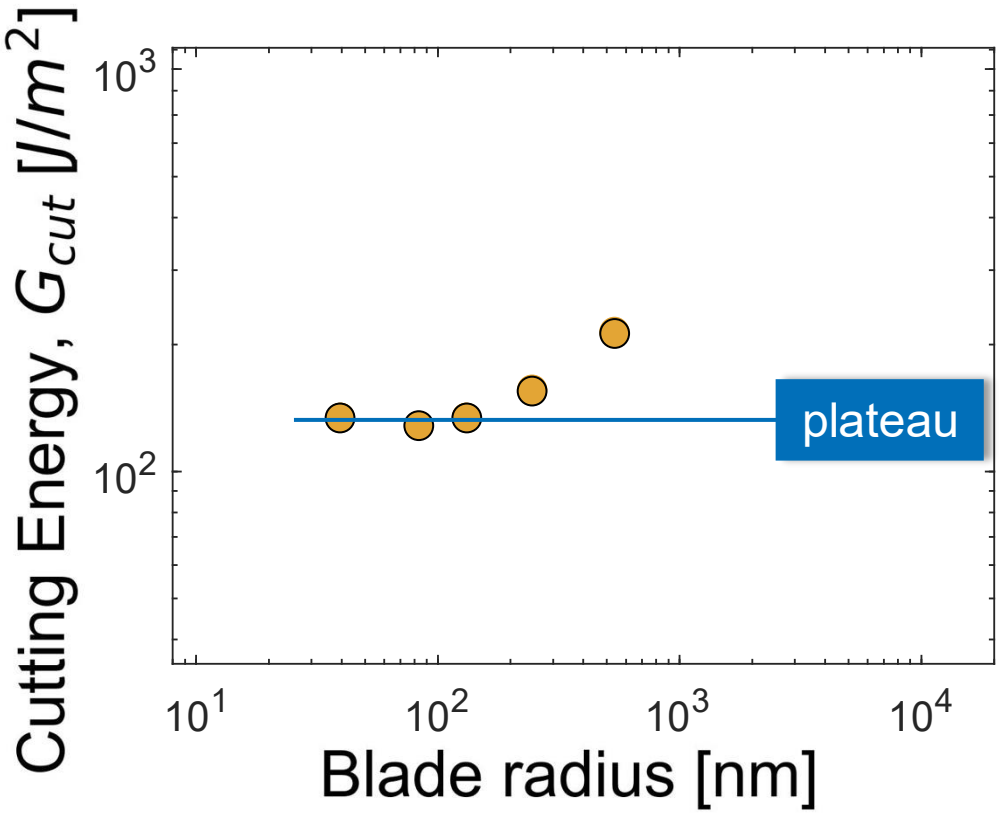
Mechanical Science and Engineering

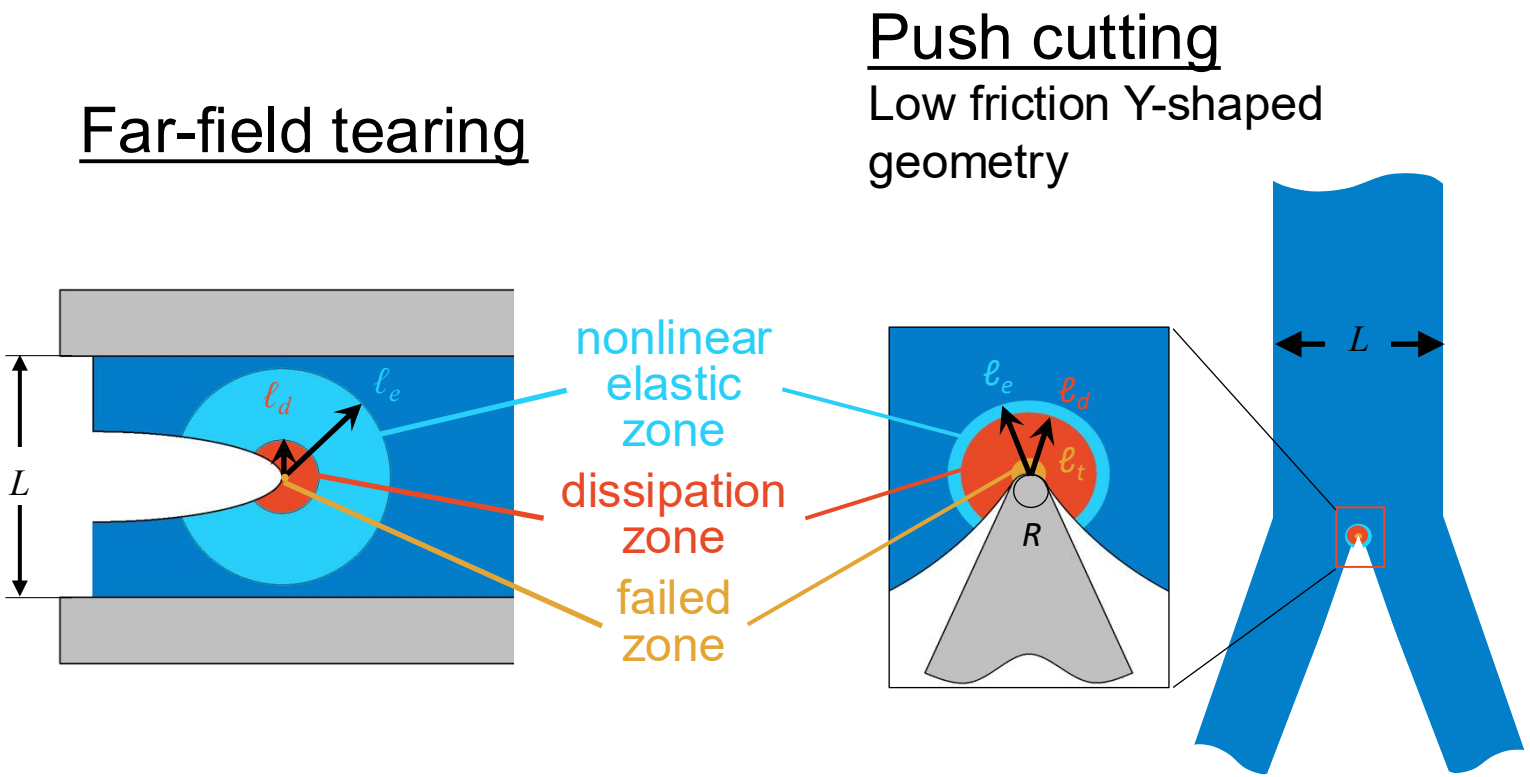
All extending cracks are not created equally.

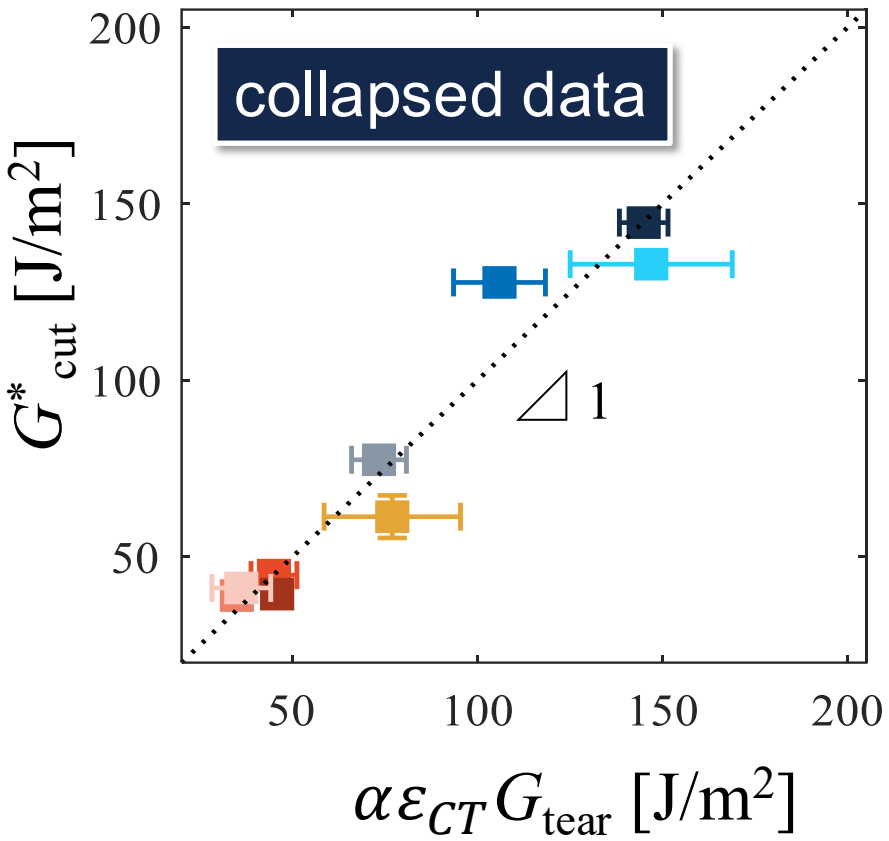




Ruobing Bai. Mechanics and Physics of Solids. (2019)
Muthukumar. Soft Matter. (2022)
Gong. Adv. Matter. (2003)
B. Zhang and S. B. Hutchens. Soft Matter. (2021)

Earlier work: Links frictionless push cutting and tearing in silicone.





- Described cutting energy radius dependence
- Connected failures operating over different length scales
- Suggested a new dimensionless group, ε_{CT}.

$$\varepsilon_{CT} \equiv \sqrt{\frac{w_b}{\mu}} \left(\frac{w_b}{\sigma_b(\lambda_b - 1)} \right)^2$$

B. Zhang and S. B. Hutchens. Soft Matter. (2021)
Long, et al. Ann. Rev. of Cond. Matter Phys. (2021)

Silicones are not the only material of interest.

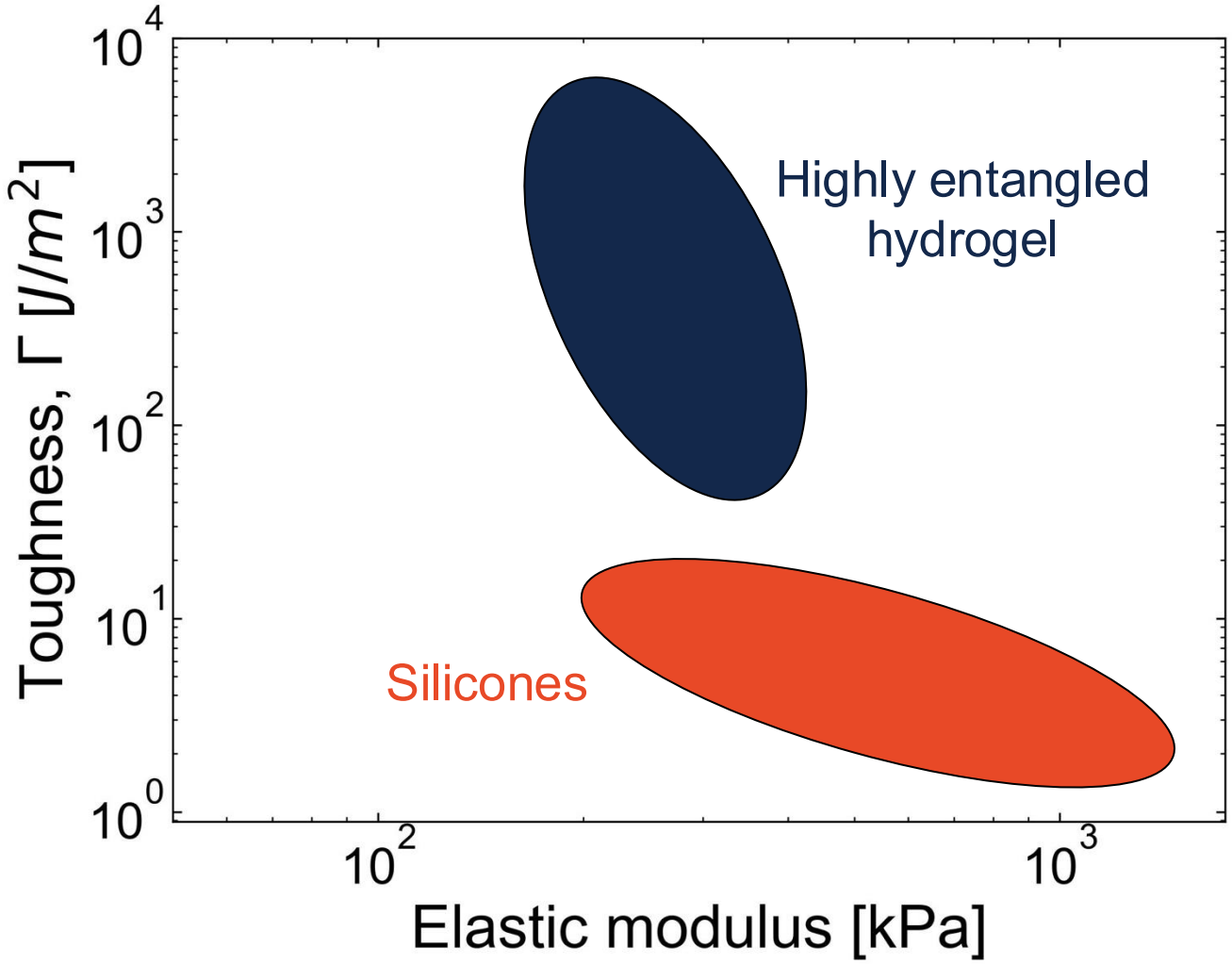
Silicones Pros:

- ✓ highly elastic
- ✓ neither too brittle nor too tough

Tough hydrogels:

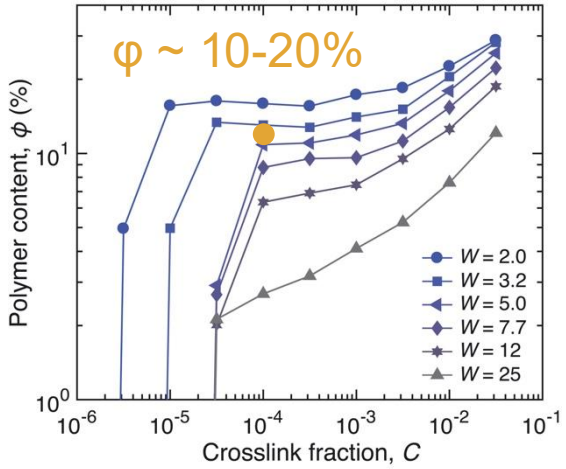
- ✓ highly elastic
- ✓ really tough

'tougher' to characterize

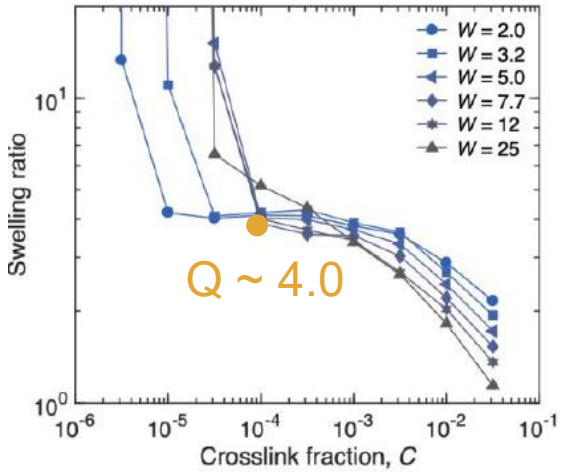


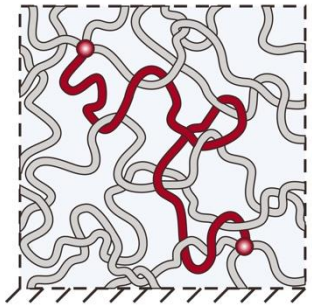
Junsoo Kim et al. Science 374,212-216 (2021)
B. Zhang and S. B. Hutchens. Soft Matter. (2021)

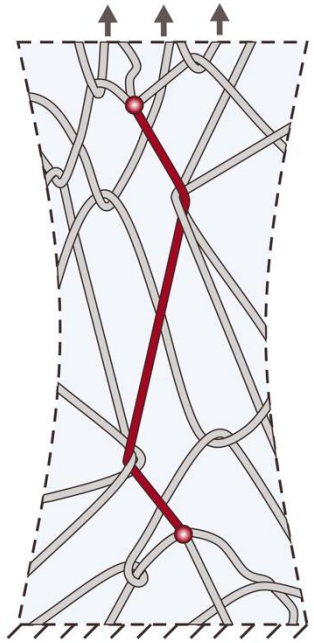
Highly entangled hydrogel - A tough material to cut

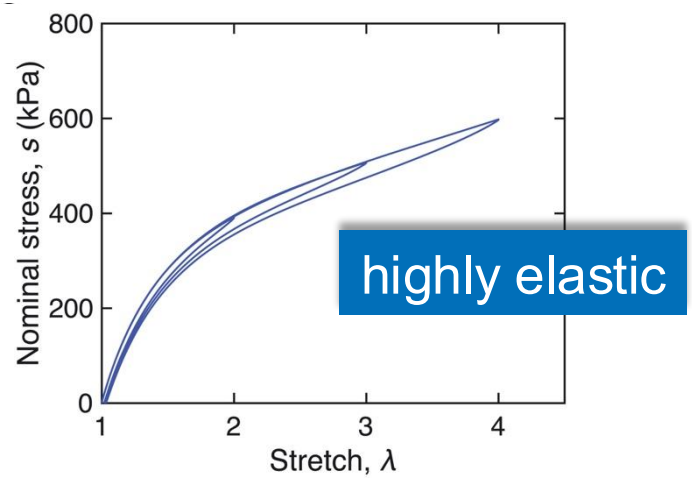


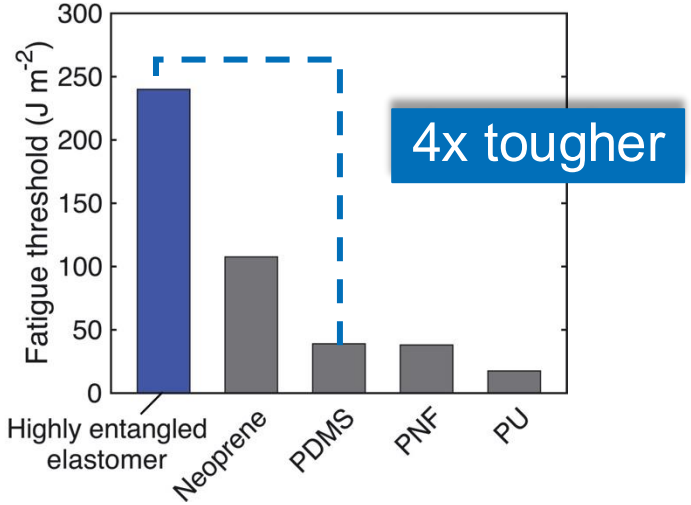
Expected to retain high water content while exhibiting high toughness





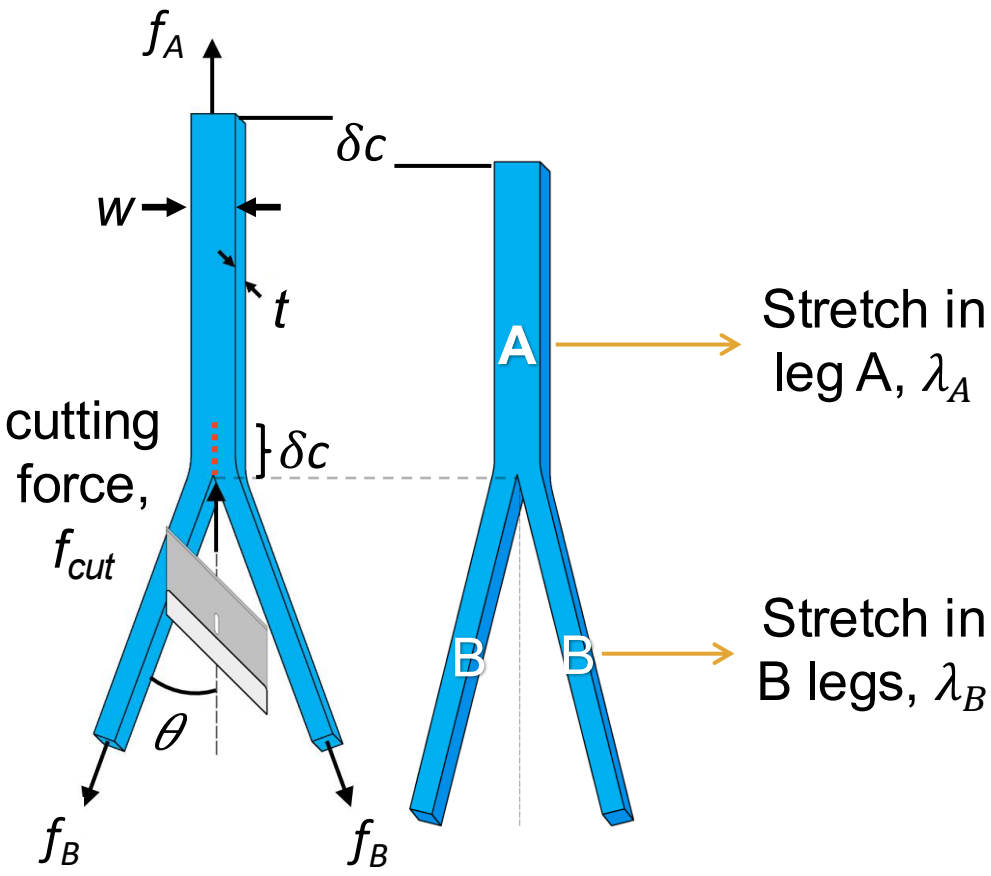


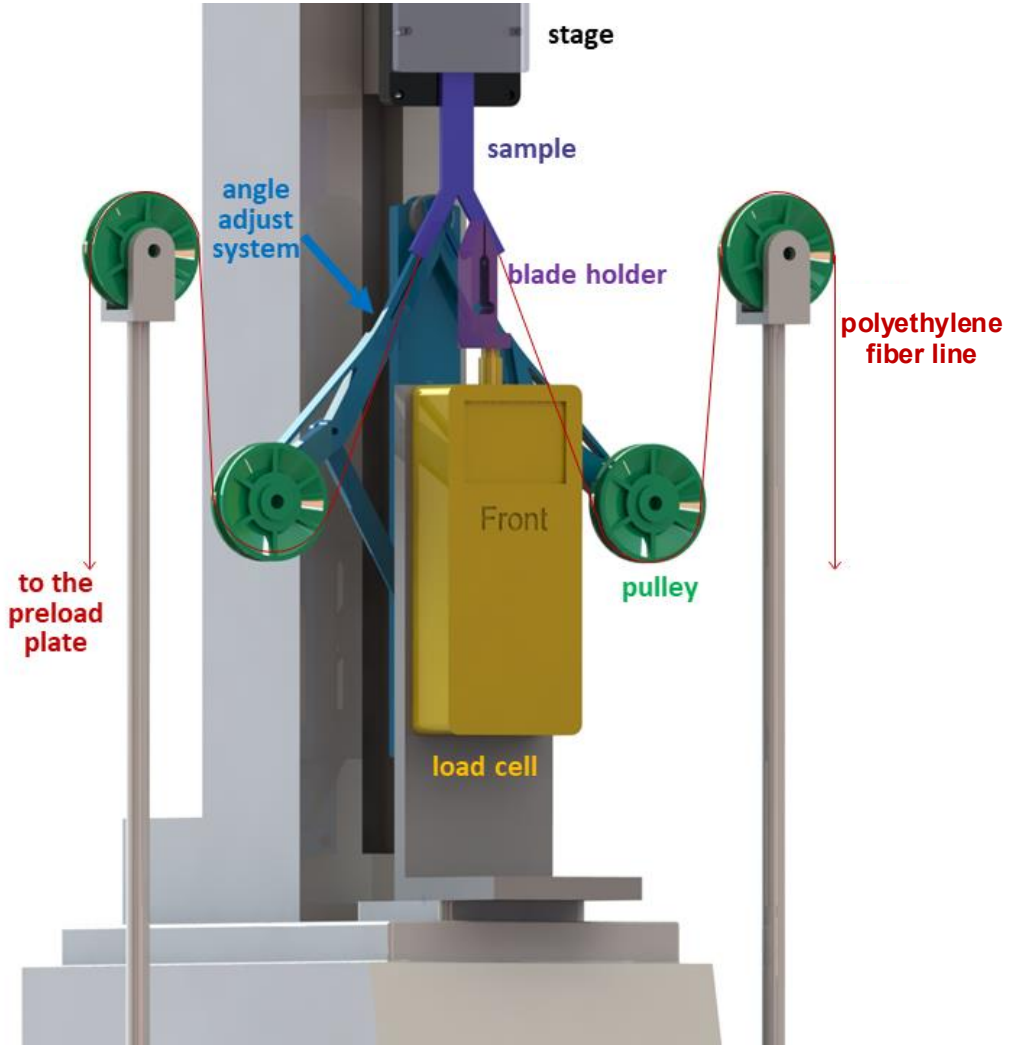




Junsoo Kim et al. Science374,212-216(2021)

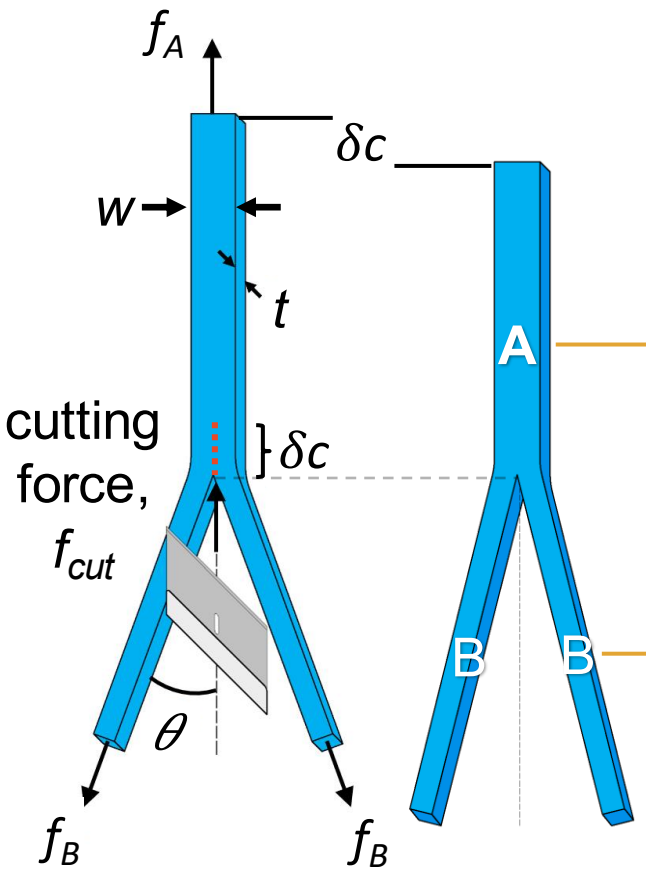
How do we do Y-shaped cutting?





Lake & Yeoh. Int J. Fract 14. (1978)
Shaobo Zhan, Hutchens lab (2023)

Strain energy release rate is simple for small leg deformation



Strain Energy Release Rate, $G = \frac{\partial U_w}{\partial A} - \frac{\partial U_e}{\partial A}$

Stretch in A legs, λ_A

Cutting Energy Release Rate

linear approximation

$$G_{cut} \approx \underbrace{\frac{f_A}{t}(\lambda_A + \lambda_B)(1 - \cos\theta)}_{\text{Tearing, } T} + \underbrace{\frac{f_{cut}}{2t}(\lambda_A + \lambda_B)}_{\text{Cutting, } C}$$

B legs, λ_B

Static equilibrium,

$$f_A = 2 f_B \cos\theta - f_{cut}$$

By linear elasticity,

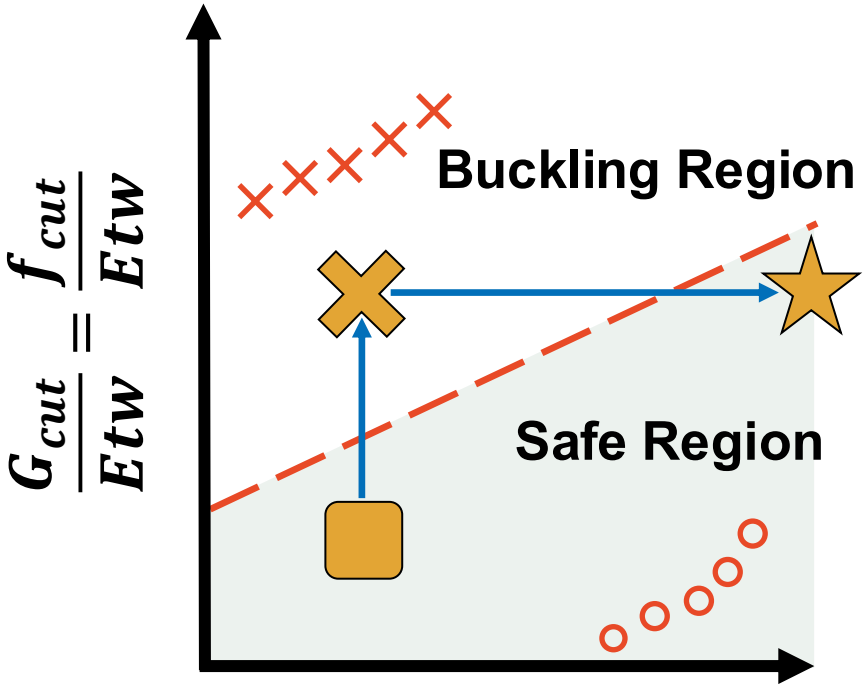
$$W = \frac{1}{2}\frac{f}{wt}(\lambda - 1)$$

Small linear uniaxial deformation:

$$\lambda_A \approx \lambda_B \approx 1$$

Lake & Yeoh. Int J. Fract 14. (1978)

Tough materials tend to buckle… but we can stabilize cutting.

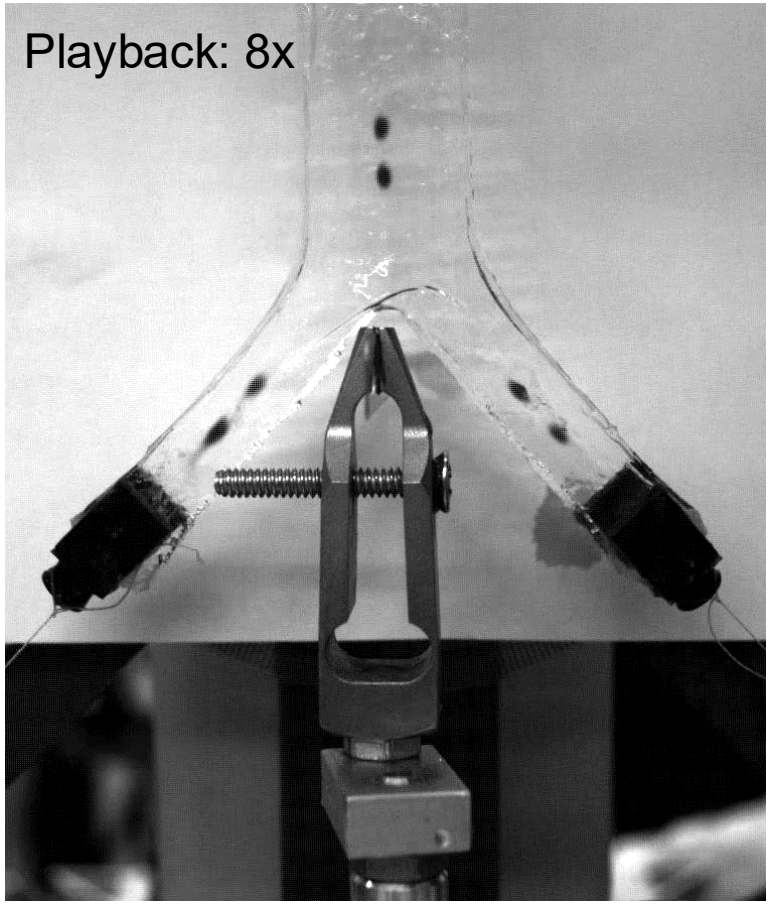


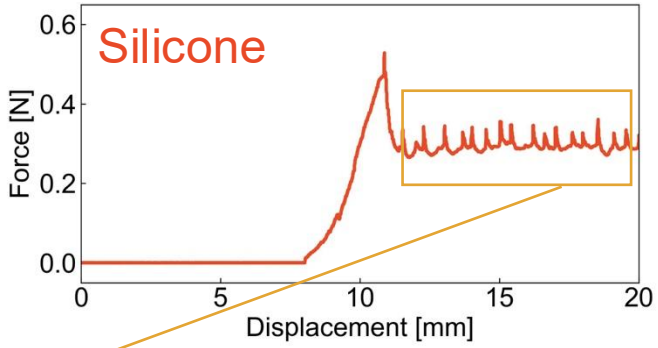
$$\frac{G_{cut}}{Etw} = \frac{f_{cut}}{Etw}$$

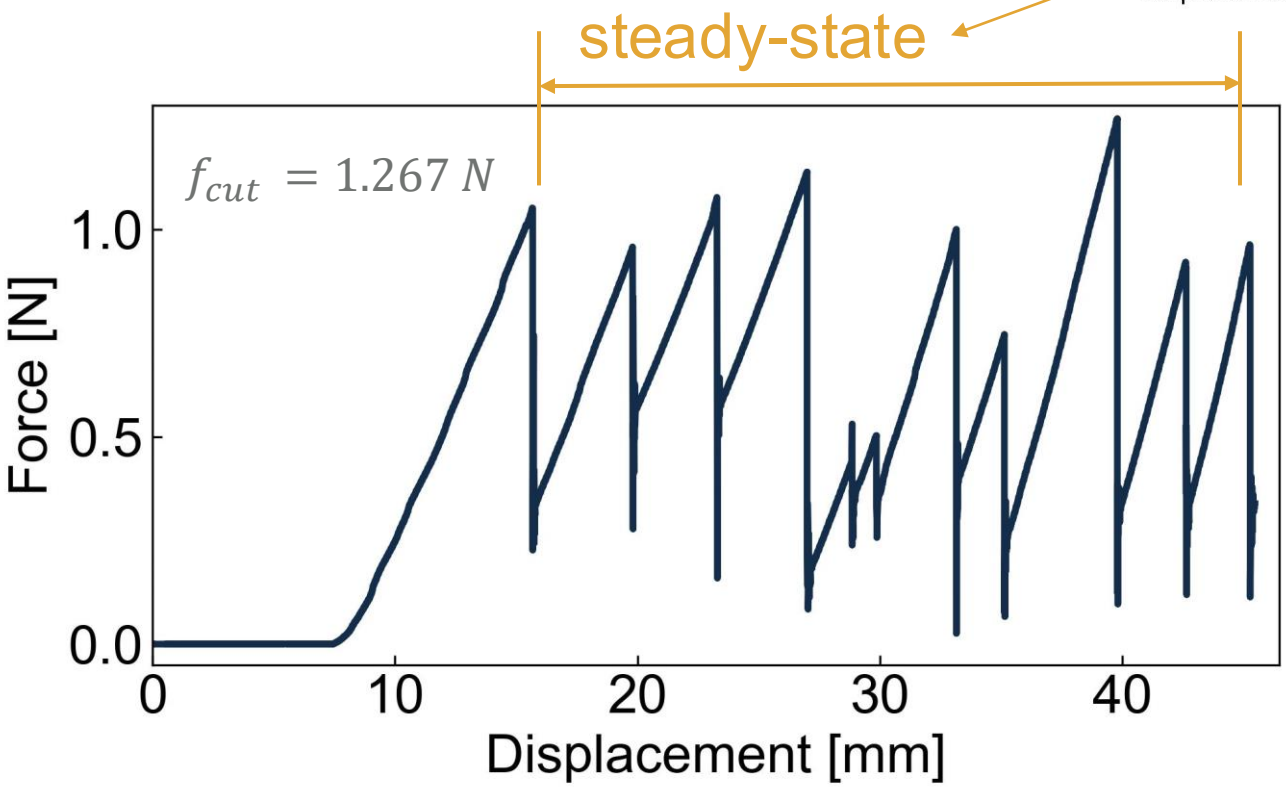
$$\frac{T}{Etw} = \frac{f_A \cos\theta}{Etw}$$

Shaobo Zhan, Hutchens lab (2023)

Yay! We successfully cut tough hydrogel



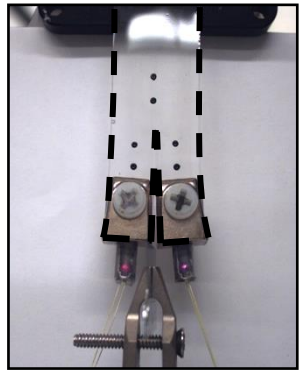


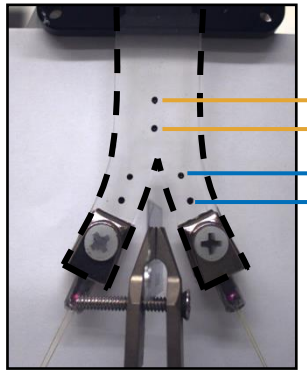


What else is affected by high preload?

Unloaded | Pre-loaded | Stretch, λ

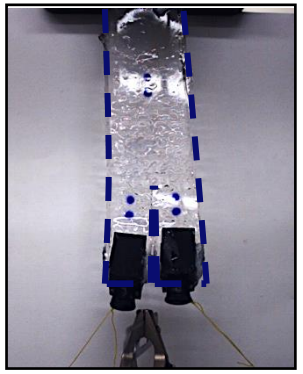
Silicone

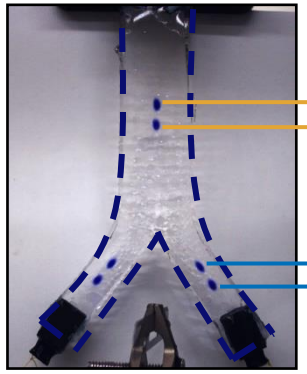




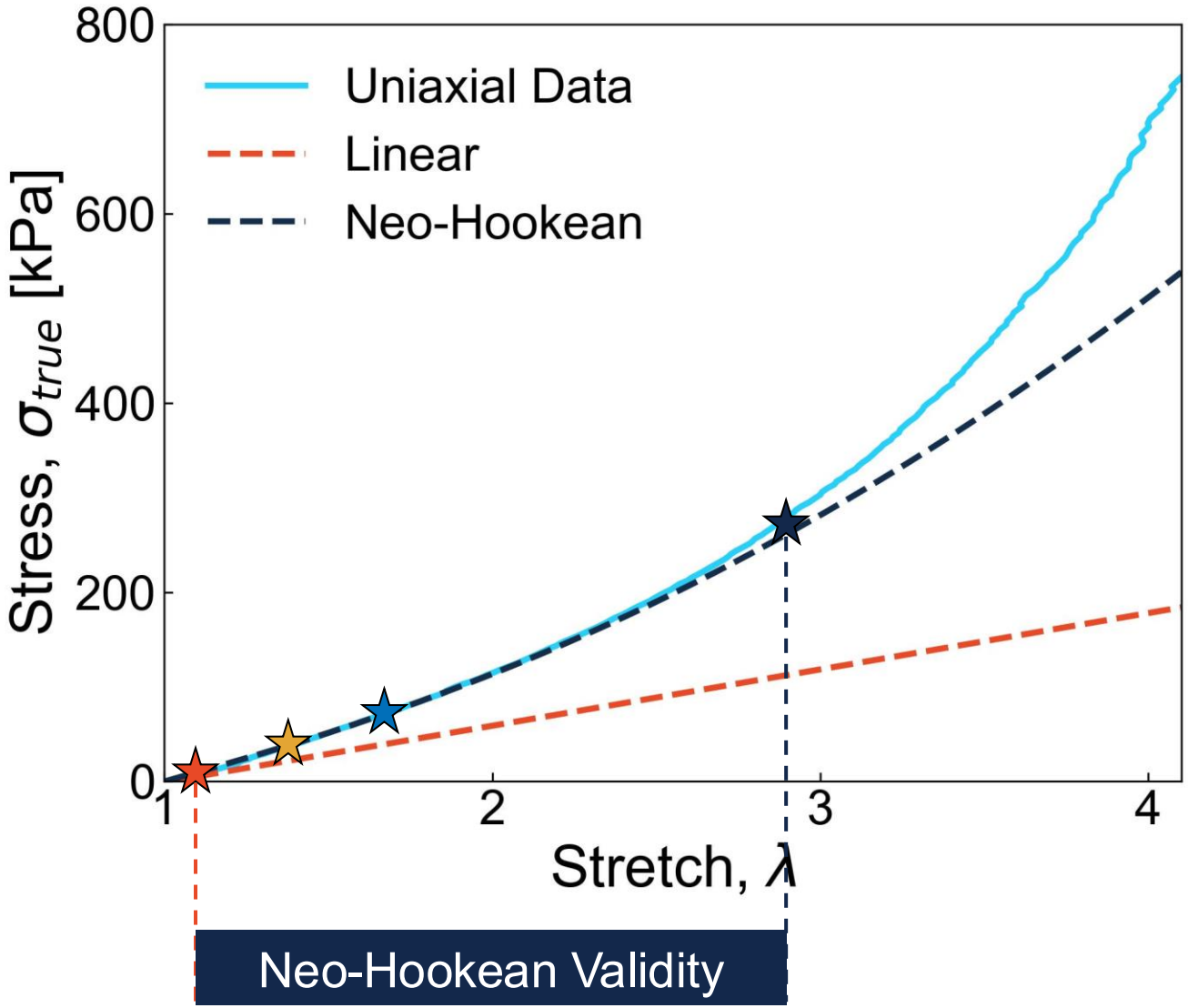
$\lambda \simeq 1$

Hydrogel

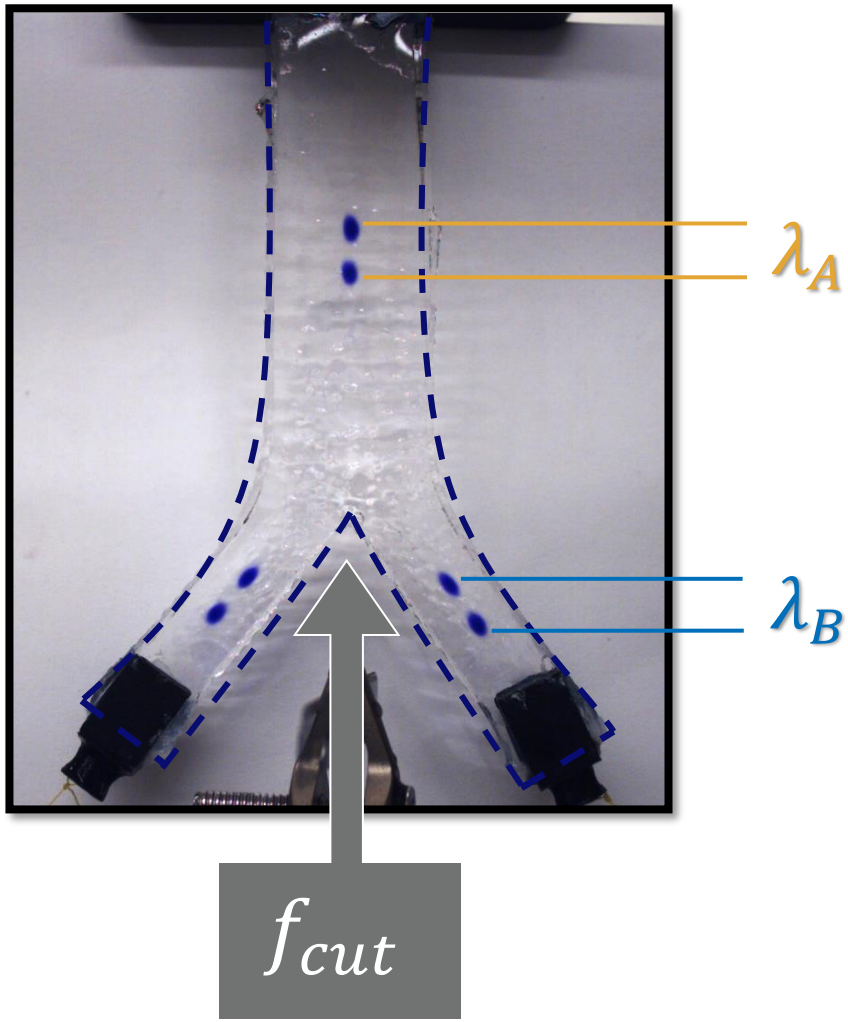


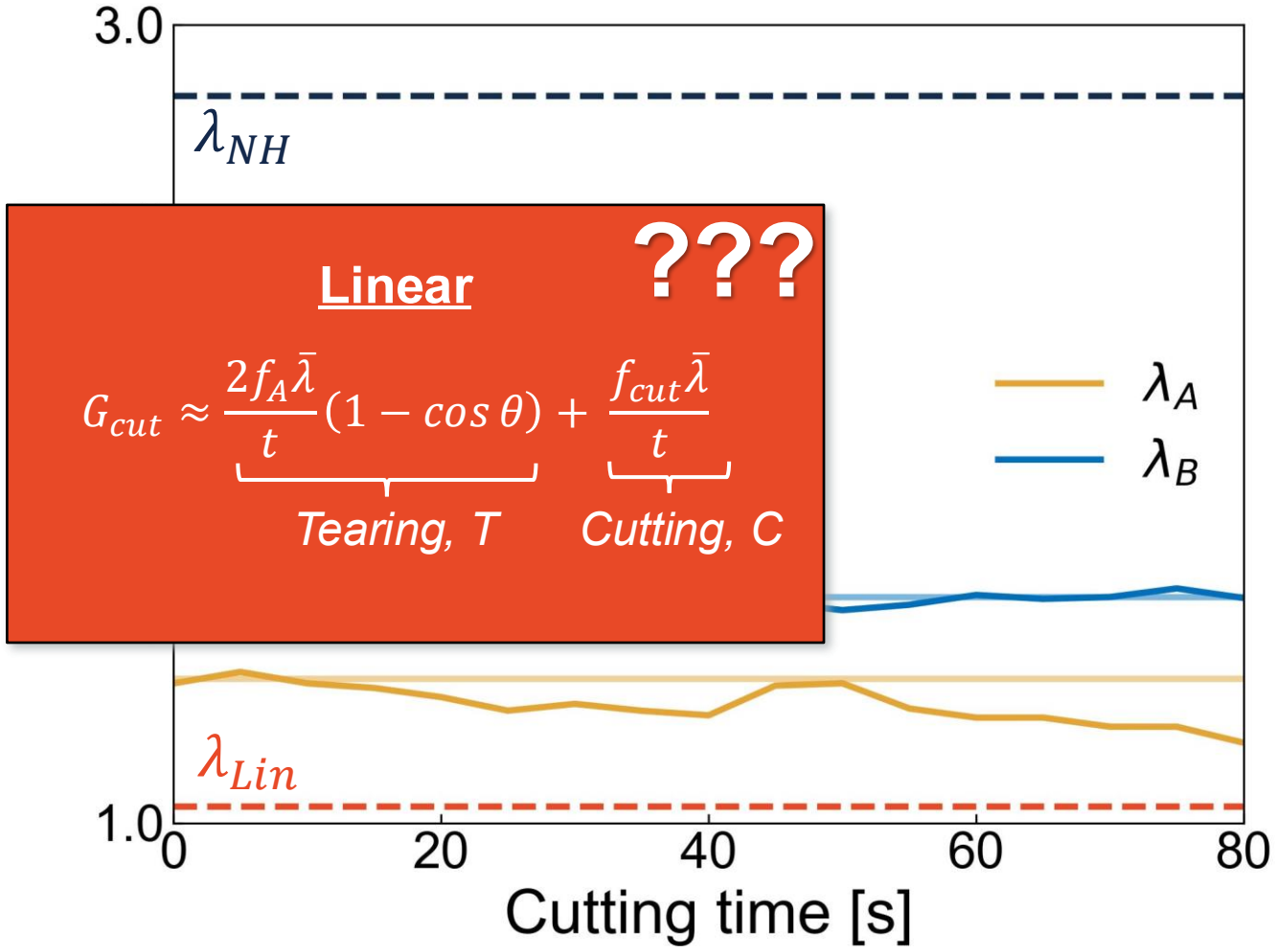


$\lambda \gtrsim 1.1$



Leg stretch must be determined during testing





Linear ???

$$G_{cut} \approx \underbrace{\frac{2 f_A \bar{\lambda}}{t}(1 - \cos\theta)}_{\text{Tearing, } T} + \underbrace{\frac{f_{cut}\bar{\lambda}}{t}}_{\text{Cutting, } C}$$

We found a non-linear G_{cut} for highly deforming materials

Linear, Small Deformation

- $\lambda_A \approx \lambda_B \approx 1$

Uniaxial deformation in the legs,

- $W = \frac{1}{2}\frac{f}{wt}(\lambda - 1)$

Linear Deformation

- $\lambda_A \neq \lambda_B \approx$ linear
- $W = \frac{1}{2}\frac{f}{wt}(\lambda - 1)$

Nonlinear Deformation

- $\lambda_A \neq \lambda_B \approx$ finite

Applying Neo-Hookean for uniaxial extension,

- $W = \frac{1}{2}\frac{f}{wt}\lambda\left(\frac{\lambda^3 - 3\lambda + 2}{\lambda^3 - 1}\right)$

Cutting Energy Release Rate

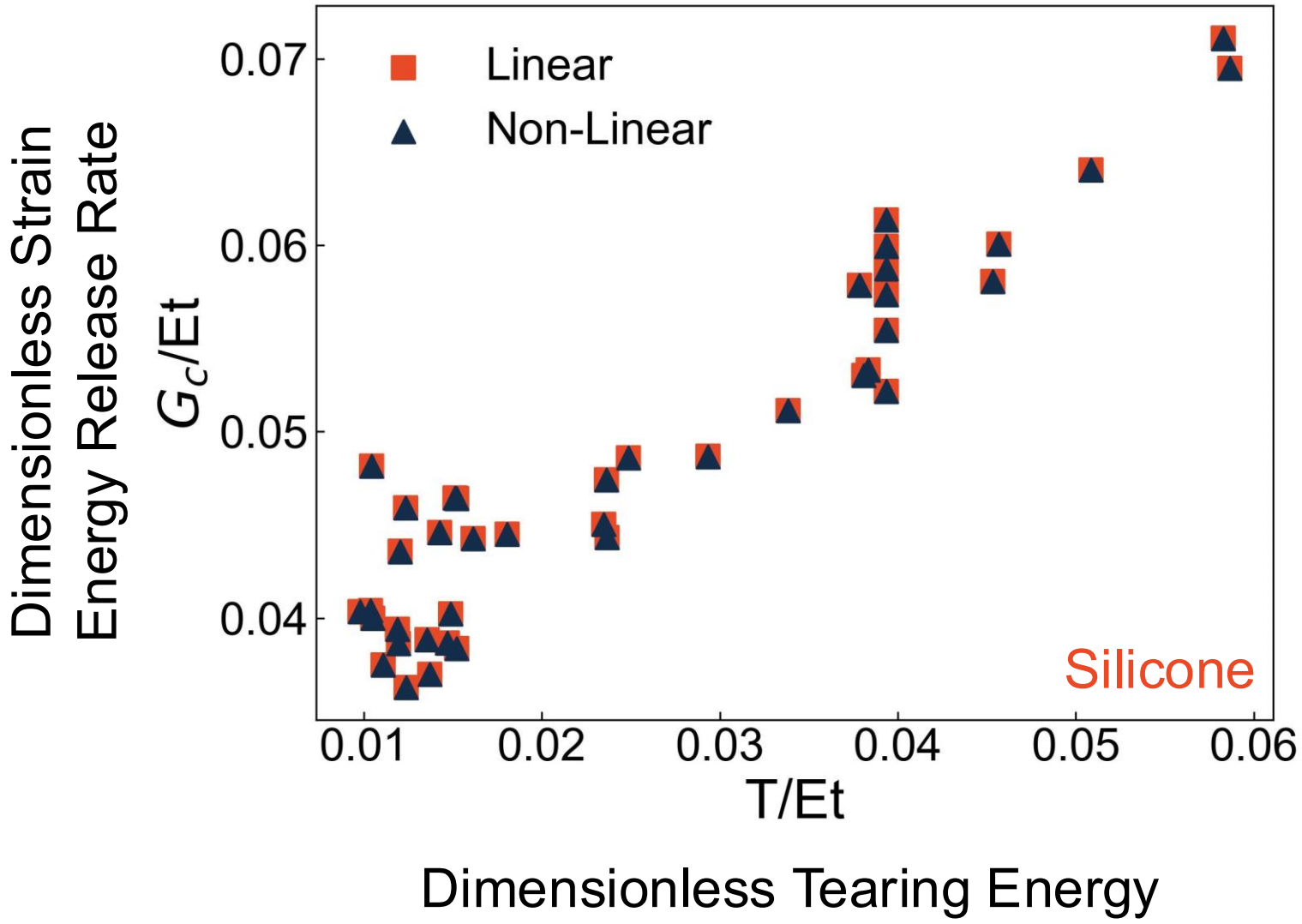
Nonlinear form

$$G_{cut} \approx \frac{2f_B\sqrt{\lambda_B}}{t}\left[\lambda_B\left(1 - \frac{\lambda_B^3 - 3\lambda_B + 2}{2(\lambda_B^3 - 1)}\right) - \lambda_A\left(1 - \frac{\lambda_A^3 - 3\lambda_A + 2}{2(\lambda_A^3 - 1)}\right)\cos\theta\right] + \frac{f_{cut}\lambda_A\sqrt{\lambda_B}}{t}\left(1 - \frac{\lambda_A^3 - 3\lambda_A + 2}{2(\lambda_A^3 - 1)}\right)$$

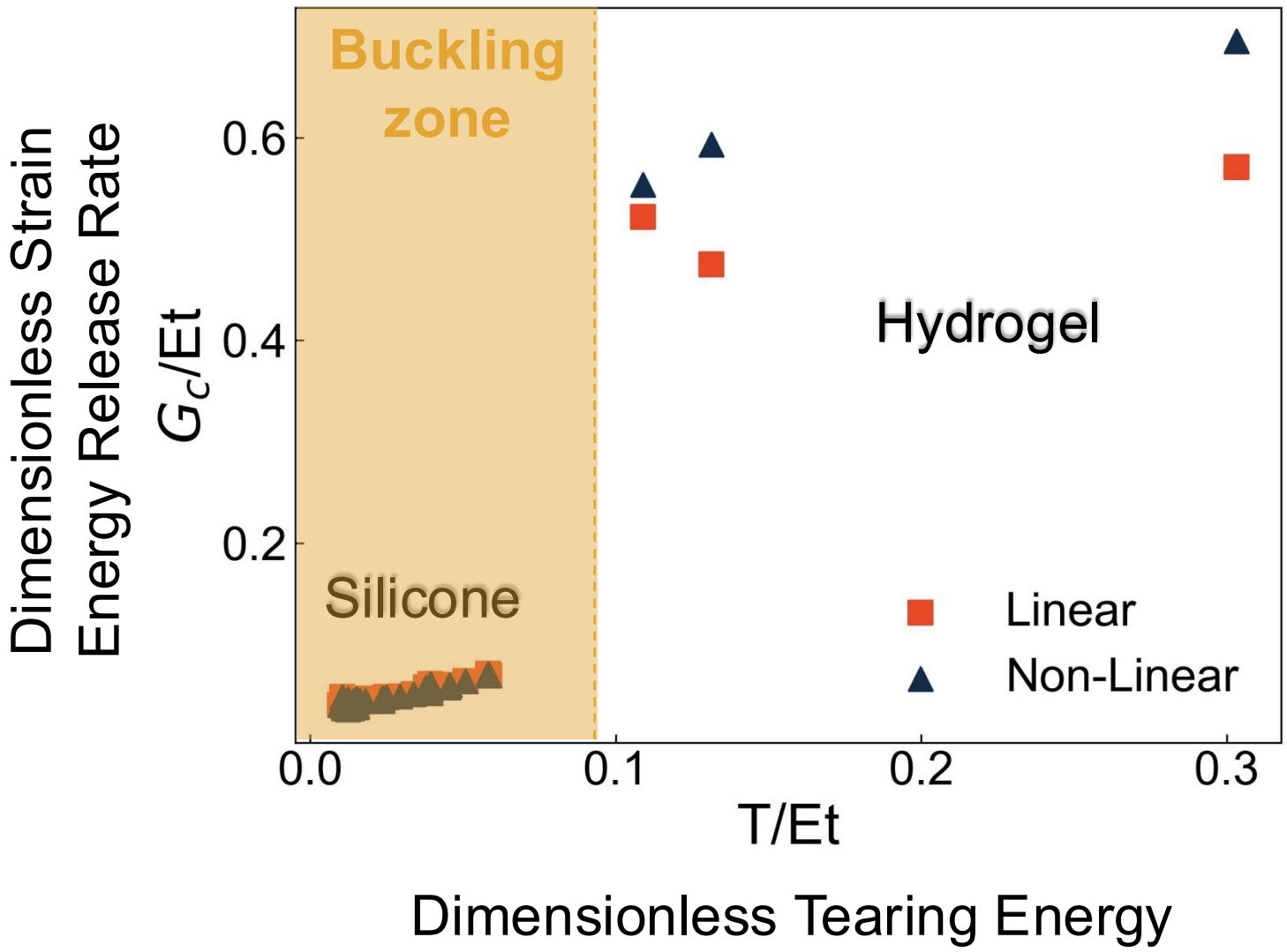
Linear approximation is recovered

in the case of:

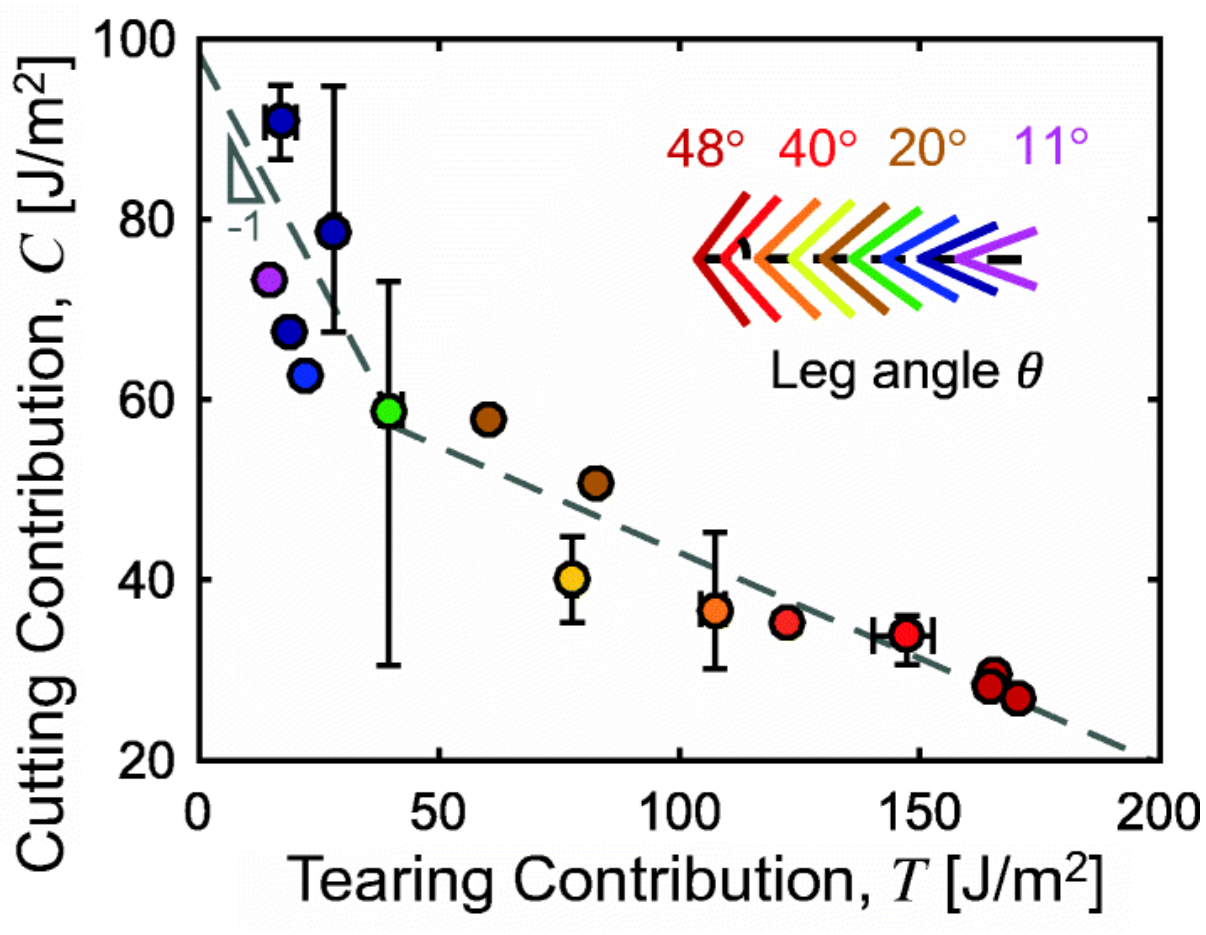
- ❑ low toughness
- ❑ small deformation



Observed up to 20% error for our materials



Next step… Is pure cutting possible?



Linear

$$G_{cut} \approx \underbrace{\frac{2 f_A \bar{\lambda}}{t}(1 - \cos\theta)}_{\text{Tearing, } T} + \underbrace{\frac{f_{cut}\bar{\lambda}}{t}}_{\text{Cutting, } C}$$

Nonlinear

$$G_{cut} \approx \frac{2 f_B \sqrt{\lambda_B}}{t}\left[\lambda_B\left(1 - \frac{\lambda_B^3 - 3\lambda_B + 2}{2(\lambda_B^3 - 1)}\right) - \lambda_A\left(1 - \frac{\lambda_A^3 - 3\lambda_A + 2}{2(\lambda_A^3 - 1)}\right)\cos\theta\right] + \frac{f_{cut}\lambda_A\sqrt{\lambda_B}}{t}\left(1 - \frac{\lambda_A^3 - 3\lambda_A + 2}{2(\lambda_A^3 - 1)}\right)$$

B. Zhang and S. B. Hutchens. Soft Matter. (2021)

To cut tough hydrogels:

- The neo-Hookean model sufficiently captures leg deformation during tough hydrogel cutting.
- Leg stretch should be determined <u>during</u> the cutting test.
- Consideration of nonlinear leg stretch in G_{cut} is important for accuracy.

Thank you!

Hutchens Lab
NIFF team

@squishiosity
Dunn & Hutchens
Research Group